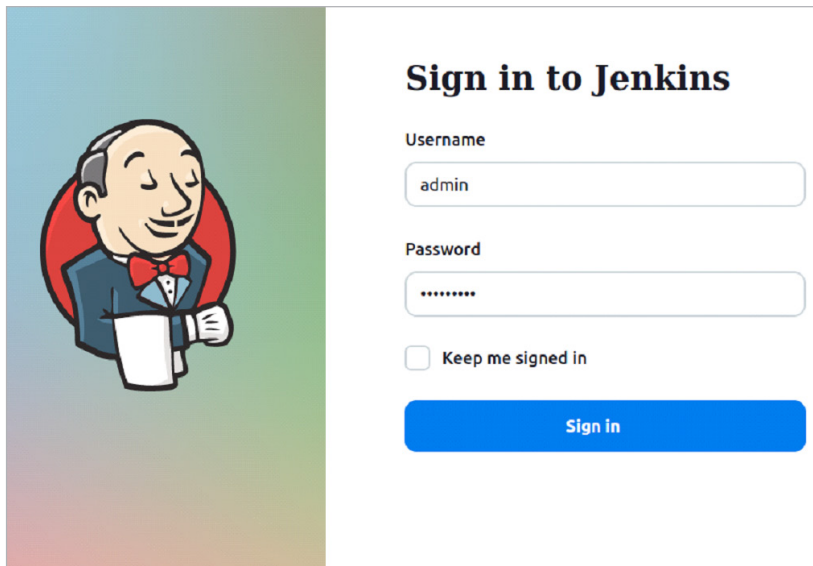




The image shows the Jenkins login interface. On the left is a cartoon illustration of a man in a suit and bow tie. On the right, the heading "Sign in to Jenkins" is displayed. Below it are fields for "Username" (containing "admin") and "Password" (masked with dots). There is a checkbox for "Keep me signed in" and a blue "Sign in" button at the bottom.

Figure 1: Jenkins login

and install it in your preferred way. I downloaded a generic war file and executed the `java -jar jenkins.war` command to run the application.

Once Jenkins is ready in your local system, visit it at `http://localhost:8080/`.

Go to *Manage Jenkins* -> *Tools* -> *Maven installations* and configure the Maven setup as shown in Figure 1.

Go to *Manage Jenkins* -> *Plugins* at `http://localhost:8080/manage/pluginManager/available` and install the Maven integration and file system SCM plugin.

Once the plugins are installed, we can create a Maven project where we can provide a local directory in the source code management section as given in Figure 3.

Go to the *Build* section in the Maven project and provide a package goal.

Go to the *Post Build* section and select *Archive Artifact* for the JAR file as given in Figure 5.

Click on *Save* and then on *Build Now*.

Verify the end section of the console output.

Verify the project dashboard with the test result chart and the downloadable JAR file.

Now, we have a JAR file ready. The next step is to create a Docker image that contains the JAR file and runtime environment.

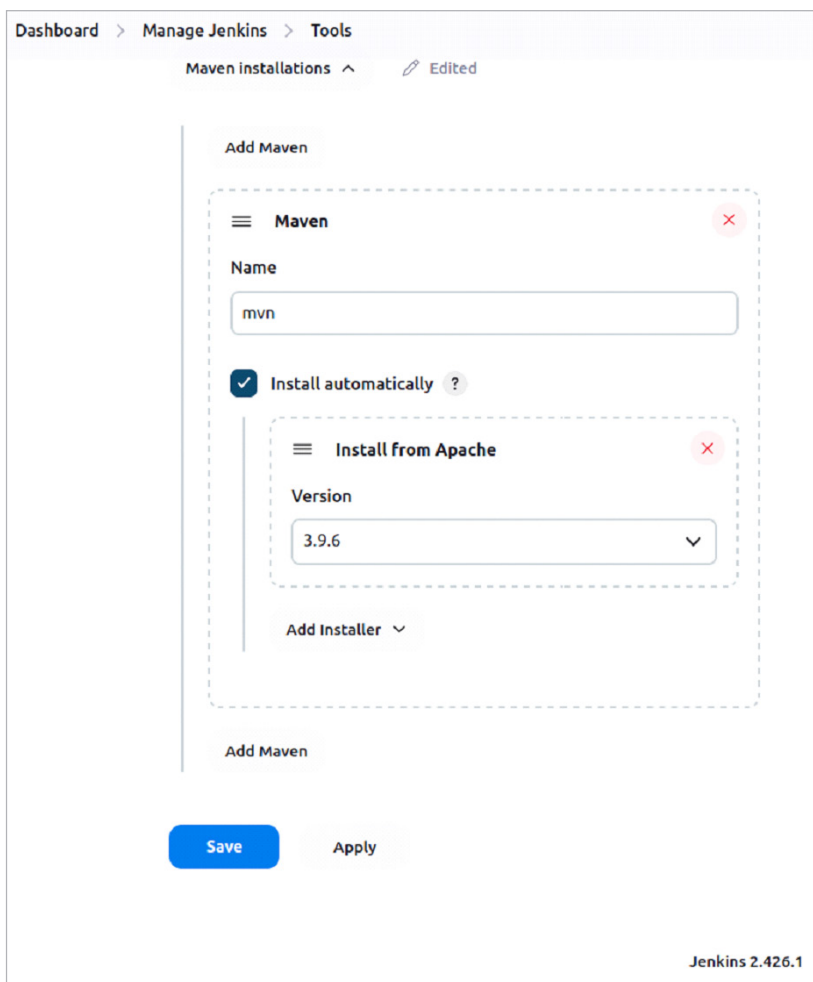
Docker installation on Ubuntu

Update the APT package index and install packages using the following command:

```
test@ubuntu:~/Desktop$ sudo apt-get update
```

```
Hit:1 https://download.docker.com/linux/
ubuntu jammy InRelease
Get:2 https://dl.yarnpkg.com/debian stable
InRelease [17.1 kB]
Get:3 https://deb.nodesource.com/node_18.x
jammy InRelease [4,563 B]
Ign:4 https://pkg.jenkins.io/
debian-stable binary/ InRelease
```

```
Get:5 https://pkg.jenkins.io/debian-stable
binary/ Release [2,044 B]
```



The image shows the Jenkins "Maven installations" configuration page. The breadcrumb trail is "Dashboard > Manage Jenkins > Tools". The page title is "Maven installations" with an "Edited" status. There is an "Add Maven" button at the top. A dashed box contains the configuration for a Maven installation named "mvn". It has a checkbox for "Install automatically" which is checked. Below it, there is a section for "Install from Apache" with a version dropdown set to "3.9.6". At the bottom of the dashed box is an "Add Installer" dropdown. Below the dashed box is another "Add Maven" button. At the bottom of the page are "Save" and "Apply" buttons. The Jenkins version "Jenkins 2.426.1" is shown in the bottom right corner.

Figure 2: Maven setup